

Topics covered: Right-angled trigonometry

The trig ratios

In a right-angled triangle the ratios relate an angle to two sides.

- $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$
- $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$
- $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$

Formulas

Name	Formula	Note
Sine ratio	$\sin(\theta) = \frac{\text{opp}}{\text{hyp}}$	Opposite over hypotenuse.
Pythagoras	$c^2 = a^2 + b^2$	Relates the three sides.

Definitions

Term	Definition
Hypotenuse	The side opposite the right angle.

Quick Reference

Concept	Summary
SOHCAHTOA	Mnemonic for the three ratios.
Exact value	$\tan(45^\circ) = 1$, $\sin(30^\circ) = \frac{1}{2}$.